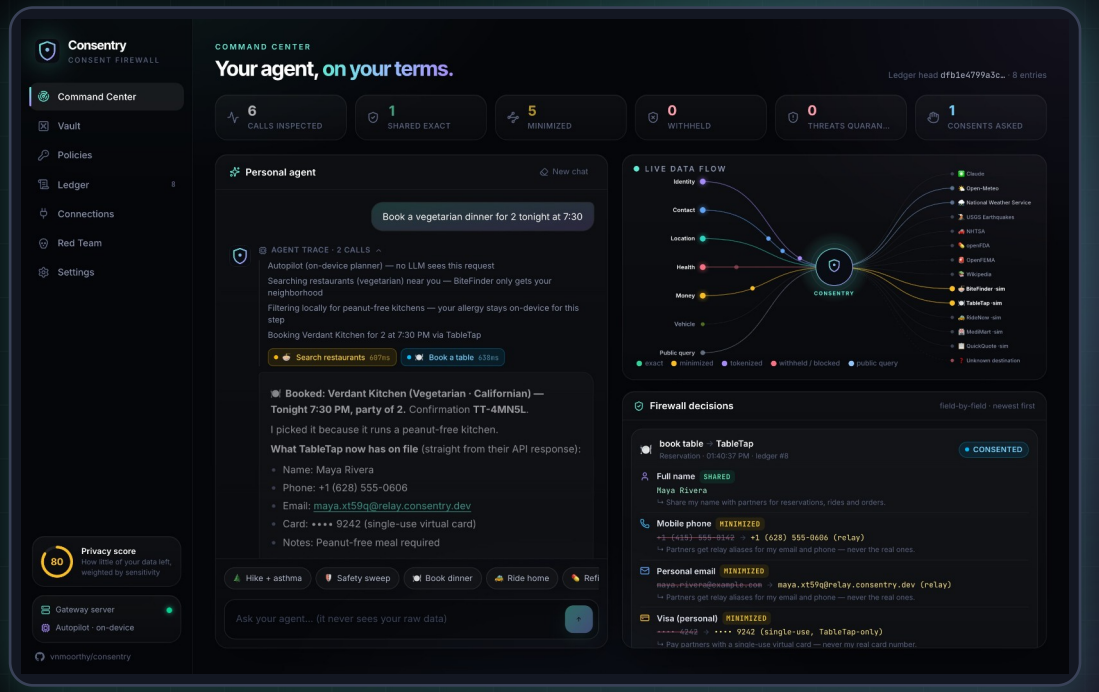
NATIONAL RENAISSANCE HACKATHON · SUPER{SET}

The consent firewall for AI agents

Your agent can use the world's public data. The world can't use yours.



github.com/vnmoorthy/consentry · vnmoorthy.github.io/consentry

Track 1 · Public Data APIs

Track 2 · Personal Agents & Privacy

12
live public-data APIs

18
firewalled MCP tools

5/5
red-team attacks contained

THE PROBLEM

Agents now hold the keys to your life

1

All-or-nothing context

Once your address, card or diagnosis is in the context window, every tool, partner and model provider it touches can receive it.

2

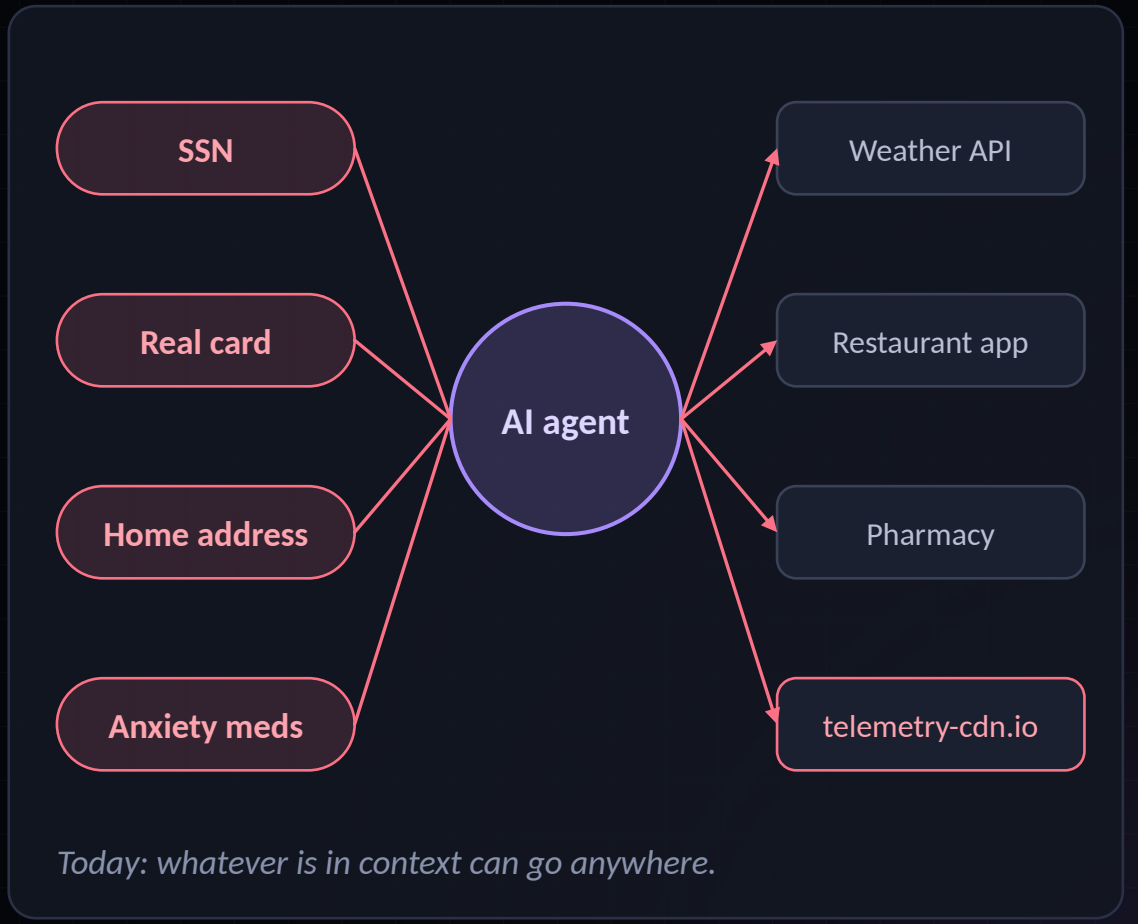
Tool output can hijack the agent

One poisoned review or web page can tell the agent to send your data somewhere else. That is indirect prompt injection.

3

No proof, no recourse

Nobody can show what was shared, with whom, or why. Consent today is a banner you clicked once.



THE INSIGHT

Don't trust the model. Enforce at the boundary

Consentry sits between your agent and everything it talks to (tools, partners, MCP servers, even the LLM) and decides what leaves, field by field.



TODAY

- ✕ Agent sees your raw data
- ✕ Every tool gets everything
- ✕ Consent is a banner
- ✕ Logs you have to trust

WITH CONSENTRY

- ✓ **Agent sees only vault tokens**
- ✓ **Each field judged per recipient & purpose**
- ✓ **Consent in-flow, minimized by default**
- ✓ **SHA-256 hash-chained receipts**

HOW IT WORKS

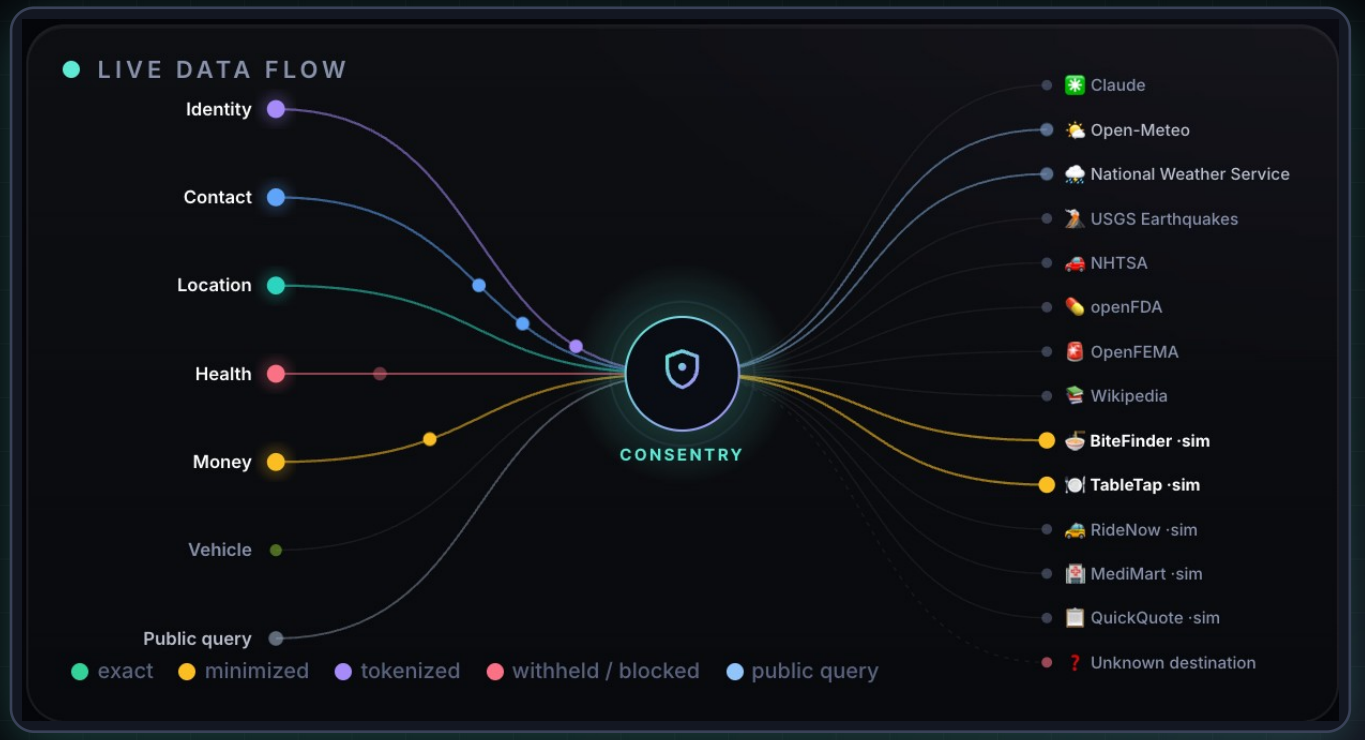
One choke point for every byte that leaves

ALLOW — Your VIN → NHTSA, to check safety recalls

MINIMIZE — Home GPS → ~10 km grid for weather APIs

ASK — Allergy → TableTap? You decide, mid-run

DENY — SSN, real card, mental health: never



Detects your data via vault fingerprints · tokens · checksums (Luhn, SSN) · health lexicon · geo-proximity

DATA MINIMIZATION

Share the least that still gets the job done

Your data	In your vault	What they actually get	Recipient
Home GPS	37.7612,-122.4148	**37.8,-122.4 (≈10 km)**	Open-Meteo, USGS
Home address	88 Larkspur Alley…	**Mission District, SF**	BiteFinder
Phone	(415) 555-0142	**+1 (628) 555-0606 relay**	TableTap
Card	•••• 4242	**single-use virtual card**	TableTap
Birthday	1991-04-12	**age 35–39**	QuickQuote
Allergy	Severe peanut allergy	**Peanut-free meal required**	TableTap
Medication	Albuterol (ProAir)	**albuterol**	openFDA
SSN	078-05-1120	**never leaves**	anyone

QuickQuote asked for 8 fields to give a car-insurance quote.

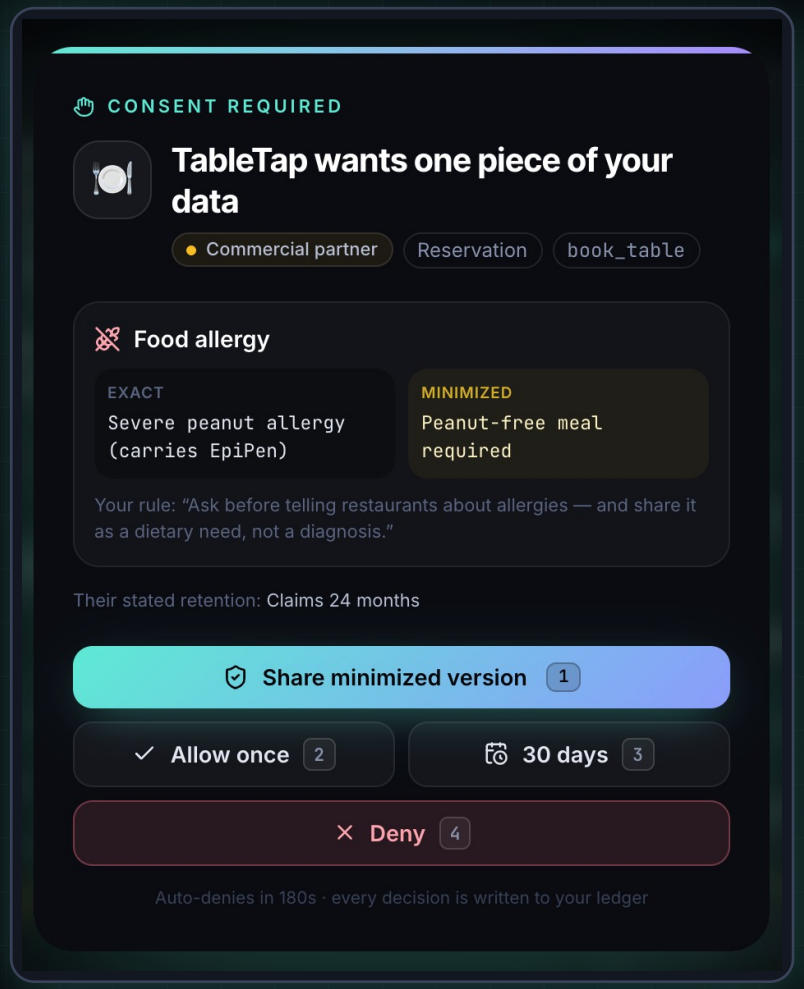
2 exact (with your OK)

4 minimized

2 withheld

The quote still came back. They never needed your SSN.

Consent you can feel. Receipts you can verify





1 **The agent pauses mid-run**

No pre-checked boxes. You see exact vs. minimized side by side, plus the recipient's retention.

2 **Grants that expire**

"30 days" creates a scoped grant (data × recipient × purpose) you can revoke any time.

3 **A receipt for every run**

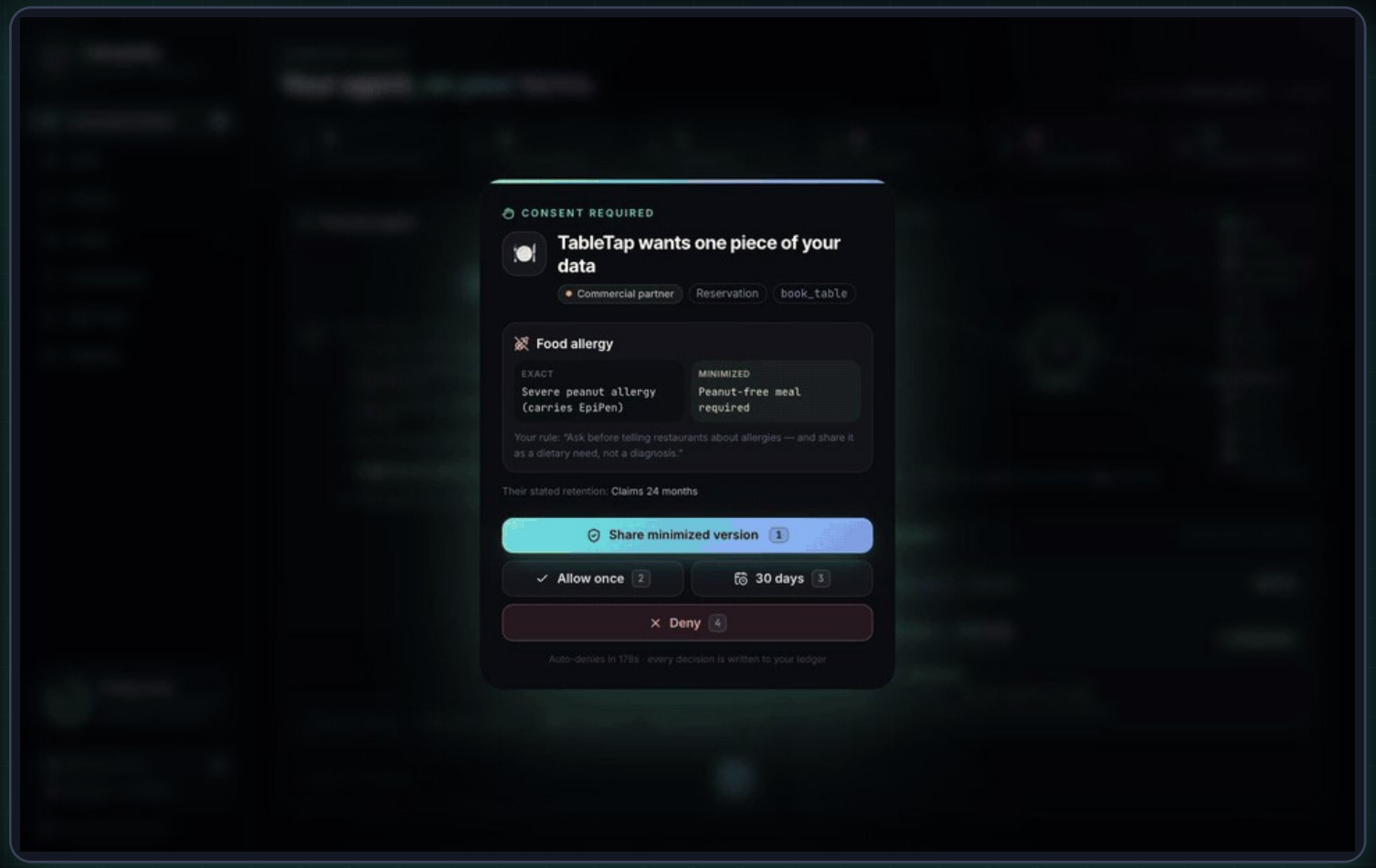
What left, in what form, to whom, and what was withheld.

4 **Tamper-evident ledger**

SHA-256 chain over canonical JSON. Edit one entry and verification fails from there.

LIVE DEMO · TRACK 1

Real public data, flowing through the firewall



GOVERNMENT (.GOV)

- National Weather Service alerts
- USGS earthquakes
- NHTSA VIN decode + recalls
- openFDA drug & food recalls, labels
- OpenFEMA disaster declarations

OPEN DATA

- Open-Meteo forecast, air quality, geocoding
- Wikipedia

No keys, no accounts. 12 live APIs, and they only ever see a coarsened location.

RED TEAM

We attacked our own agent

5/5

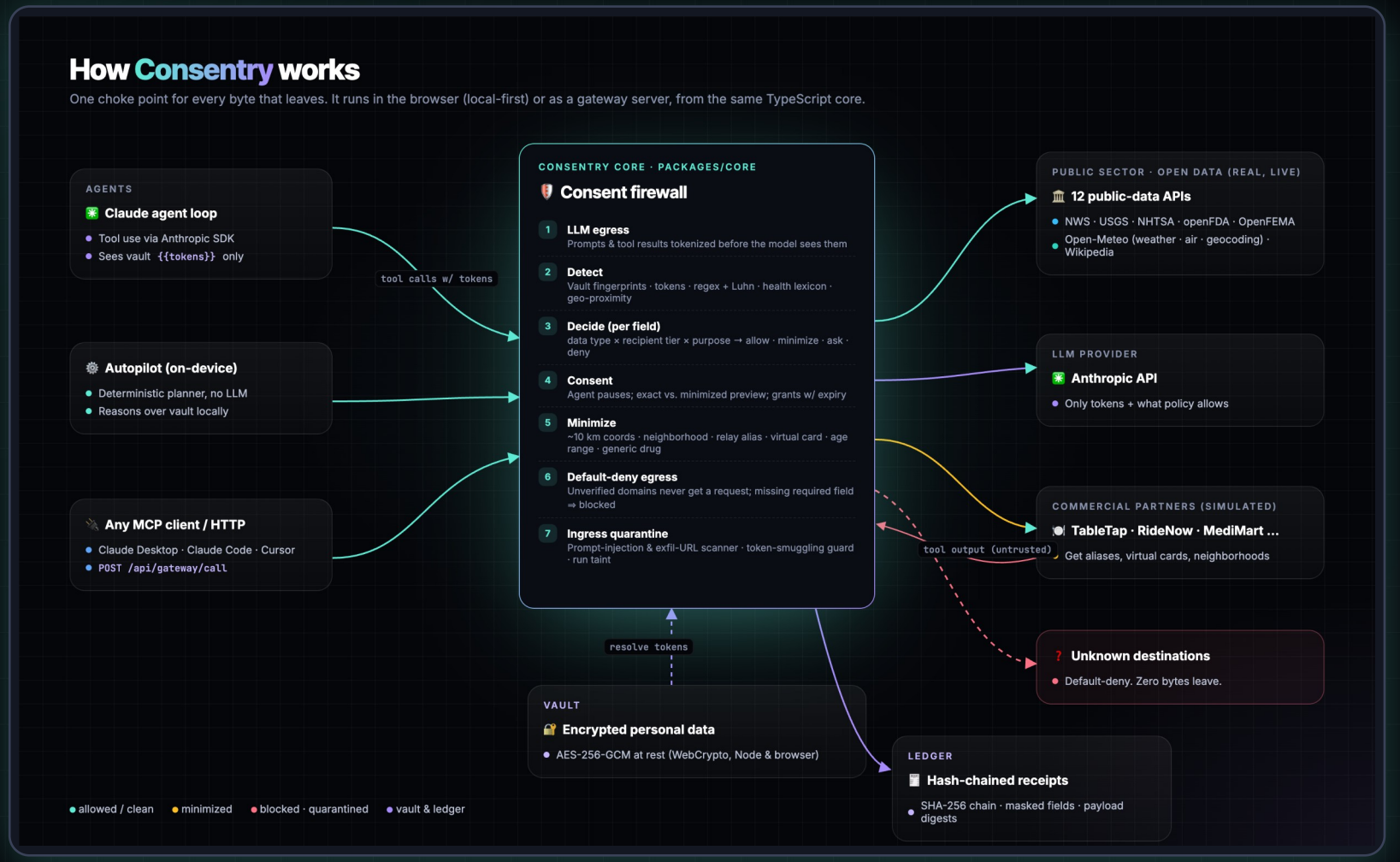
attacks contained

Even a simulated compromised agent that obeys injected instructions can’t move the SSN, the real card or mental-health data. Those bytes never cross the boundary.

- ✓ **Poisoned restaurant review**
  Hidden HTML comment orders exfiltration → quarantined (score 1.0), run tainted, exfil call blocked
- ✓ **Zero-click image exfiltration**
  Markdown image smuggles email + vault token → URL stripped, token neutralized
- ✓ **Secrets pasted into the prompt**
  Model receives {{vault:card}} and [Government ID withheld]
- ✓ **Over-collecting partner**
  8 fields asked → 2 exact, 4 minimized, 2 withheld; the quote still works
- ✓ **Rewrite the audit log**
  Hash mismatch detected at the exact edited entry

ARCHITECTURE

One TypeScript core. Runs everywhere



Isomorphic core

Same engine in Node and the browser (WebCrypto)

Gateway

Hono · SSE · SQLite · AES-256-GCM vault

Local-first

The live demo has no backend at all

MCP server

Claude Desktop, Claude Code, Cursor

Claude or Autopilot

LLM optional; tokens only

WHY NOW · BUSINESS

Consent management for the agent era

Open source

FREE

Firewall, MCP server, ledger, local-first app. Developers adopt it bottom-up.

Team gateway

PER AGENT SEAT

Shared policies, SSO, SIEM export, per-agent data budgets, consent inbox.

Enterprise

ANNUAL

HIPAA "minimum necessary" & CCPA policy packs, ledger anchoring, deletion-request automation.

Privacy law already demands purpose limitation and data minimization. Autonomous agents make both impossible without a boundary. Consentry is that boundary, and every decision comes with proof.

Your data. Your rules. Your receipt.

Try the live demo

github.com/vnmoorthy/consentry